UNIVERSITY OF CAPE TOWN

DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics II Advanced Calculus (2AC)

Double Integrals over non-rectangular domains

Dr Francois Ebobisse Bille

0.1 Definition and properties

Let D be a non-rectangular bounded domain in $\mathbb{R}^2$ and let $f : D \to \mathbb{R}$ be a continuous function.

(1) We consider a rectangle D' containing the domain D and we extend the function by zero outside the domain D as follows: we set

$$\tilde{f}(x,y) := \begin{cases} f(x,y) & \text{if } (x,y) \in D, \\ \\ 0 & \text{if } (x,y) D' \setminus D\,. \end{cases}$$

(2) Now, the concept of Riemann integrability is used to show that the Riemann integral

$$\iint_{D'} \tilde{f}(x,y)dxdy$$

of the new function $\tilde{f}$ over the rectangle D' is well-defined, independent of the rectangle D' and dependents on the function f.

(3) By definition

$$\boxed{\iint_D f(x,y)dxdy := \iint_{D'} \tilde{f}(x,y)dxdy} \tag{1}$$

Some properties of double integrals:

Let $f, g : D \subset \mathbb{R}^2 \to \mathbb{R}$ be continuous functions. The following properties hold.

(a) $\displaystyle\iint_D (\alpha f(x,y) + \beta g(x,y))dxdy = \alpha \iint_D f(x,y)dxdy + \beta \iint_D g(x,y))dxdy \quad$ for α, $\beta \in \mathbb{R}$;

(b) $\displaystyle\iint_D f(x,y)dxdy \leq \iint_D g(x,y)\,dxdy$ whenever $f(x,y) \leq g(x,y)$ for every $(x,y) \in D$;

(c) $\displaystyle\iint_D f(x,y)dxdy = \iint_{D_1} f(x,y)dxdy + \iint_{D_2} f(x,y)dxdy$ whenever $D = D_1 \cup D_2$ with D_1 and D_2 having only boundary points in common;

(d) $\displaystyle\left|\iint_D f(x,y)dxdy\right| \leq \iint_D \big|f(x,y)\big|dxdy$;

(e) If $m \leq f(x,y) \leq M$ for every $(x,y) \in D$, then $\displaystyle mA(D) \leq \iint_D f(x,y)\,dxdy \leq MA(D)$;

0.2 Evaluation of the double integral

In order to evaluate a double integral in a non-rectangular domain D one has first to examine carefully the domain D. Precisely, it requires that one identifies the boundary curves of the domain D.

Theorem 0.1 (Second Fubini's theorem)
Let $f : D \subset \mathbb{R}^2 \to \mathbb{R}$ be a continuous function.

(i) *If* $D := \{(x,y) \in \mathbb{R}^2\colon\ a \leq x \leq b,\ \ g_1(x) \leq y \leq g_2(x)\}$ *where*

$$g_1, g_2 : [a,b] \to \mathbb{R} \qquad \textit{are continuous functions such that} \qquad g_1(x) \leq g_2(x) \quad \forall x \in [a,b]$$

then,

$$\boxed{\iint_D f(x,y)dxdy = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y)dydx}\,. \tag{2}$$

(ii) *If* $D := \{(x,y) \in \mathbb{R}^2\colon\ c \leq y \leq d,\ \ h_1(y) \leq x \leq h_2(y)\}$ *where*

$$h_1,\ h_2 : [c,d] \to \mathbb{R} \qquad \textit{are continuous functions such that} \qquad h_1(y) \leq h_2(y) \quad \forall y \in [c,d]$$

then,

$$\boxed{\iint_D f(x,y)dxdy = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x,y)dxdy}\,. \tag{3}$$

The evaluation of a double integral is done through the process of successive integration for suitable integrals of functions of a single variable.

Example 0.2 Evaluate the double integral $\displaystyle\iint_D xydxdy$ where D is the domain between the curves $y = x^2$ and $x = y^2$ for $0 \leq x \leq 1$.

Solution: We are going to evaluate the double integral according to the following three steps:

(1) Find the points of intersection of the two curves $y = x^2$ and $x = y^2$. We find that the two curves intersect at the points $(0,0)$ and $(1,1)$.

(2) Sketch the domain D!

(3) We choose the other of integration in the successive integration. Quite often, the challenge in this step is to identify the correct limits of the integrals. In the present case, we choose to first integrate w.r.t. y and get

$$\begin{aligned}\iint_D xydxdy &= \int_0^1 \left(\int_{x^2}^{\sqrt{x}} xydy\right) dx = \int_0^1 \left[\frac{1}{2}xy^2\right]_{x^2}^{\sqrt{x}} dx = \frac{1}{2}\int_0^1 \left[x^2 - x^5\right] dx \\ &= \frac{1}{2}\left[\frac{1}{3}x^3 - \frac{1}{6}x^6\right]_0^1 = \frac{1}{2}\left(\frac{1}{3} - \frac{1}{6}\right) = \frac{1}{12}\,.\end{aligned}$$

If we change the order of integration i.e., we describe the domain D as

$$D = \{(x,y) \in \mathbb{R}^2\colon\ 0 \leq y \leq 1,\ y^2 \leq x \leq \sqrt{y}\}$$

we will get

$$\iint_D xydxdy = \int_0^1 \int_{y^2}^{\sqrt{y}} xydxdy = \frac{1}{2}\int_0^1 \left[y^2 - y^5\right] dy = \frac{1}{12} \quad \text{(just as before)}\,.$$

Remark 0.3 As we will see from various examples, this process involves the challenge of choosing the right order of integration. It can happen that integration can be performed in one order while it is impossible in the reversed order. For instance,

(i) let us evaluate the double integral

$$\boxed{\int_0^{\frac{\pi}{2}} \int_{\sqrt{x}}^{\sqrt{\frac{\pi}{2}}} \frac{1}{y} \cos(y^2) dy dx}. \tag{4}$$

Notice that the function $\frac{1}{y}\cos(y^2)$ has no elementary antiderivative. It is impossible to evaluate the integral in the given order through any known technique of integration. Therefore, there is a need to change the order.

To this aim, the first step is to clearly identify (by sketching) the domain D

Notice that (4) is a double integral over the domain D enclosed by the y-axis, the line $y = \sqrt{\frac{\pi}{2}}$ and the curve $x = y^2$ (**SKETCH D!!**). So, changing the order in (4), we obtain that

$$\boxed{\int_0^{\frac{\pi}{2}}\int_{\sqrt{x}}^{\sqrt{\frac{\pi}{2}}} \frac{1}{y}\cos(y^2)dydx = \int_0^{\sqrt{\frac{\pi}{2}}}\int_0^{y^2} \frac{1}{y}\cos(y^2)dxdy = \int_0^{\sqrt{\frac{\pi}{2}}} y\cos(y^2)dy = \Big[\frac{1}{2}\sin(y^2)\Big]_0^{\sqrt{\frac{\pi}{2}}} = \frac{1}{2}}. \tag{5}$$

(ii) Let us now evalute the double integral $\iint_D (x+y)dxdy$ where D is the domain bounded by the x-axis, the lines $x = -1$, $x = 1$ and the curve $y = x^2 + 1$. We we write D as

$$D := \{(x, y) \in \mathbb{R}^2 : \; -1 \le x \le 1, \; 0 \le y \le x^2 + 1\}.$$

So, we get

$$\begin{aligned}
\iint_D (x+y)dxdy &= \int_{-1}^{1}\int_0^{x^2+1} (x+y)dydx = \int_{-1}^{1} \Big[xy + \frac{1}{2}y^2\Big]_0^{x^2+1} dx \\
&= \int_{-1}^{1} \Big[x + x^3 + \frac{1}{2} + x^2 + \frac{1}{2}x^4\Big]dx = 2\int_0^1 \Big[\frac{1}{2} + x^2 + \frac{1}{2}x^4\Big]dx \\
&= 2\Big[\frac{1}{2}x + \frac{1}{3}x^3 + \frac{1}{10}x^5\Big]_0^1 = 2\Big(\frac{1}{2} + \frac{1}{3} + \frac{1}{10}\Big) = \frac{28}{15}.
\end{aligned}$$

TRY TO REVERSE THE ORDER OF INTEGRATION!!!!

Notice that the symbol $\boxed{\iint_D f(x,y)dxdy}$ **simply stands for the double integral, with no significance to whether the integration w.r.t.** x **precedes the integration w.r.t.** y.

Exercises 0.4

1. Evaluate the double integral over the given domain.

   (a) $\displaystyle\iint_D x\cos y\,dxdy$, where D is the domain enclosed by the lines $y=0$, $x=\dfrac{\pi}{4}$ and $y=2x$.
   $\boxed{\text{Answer: } \dfrac{1}{4}}$.

   (b) $\displaystyle\iint_D 2x\,dxdy$, where $D := \left\{(x,y)\in\mathbb{R}^2\colon\ x\geq 0,\ y\geq\dfrac{\sqrt{3}}{2},\ \dfrac{x^2}{4}+y^2\leq 1\right\}$.

   Hint: $\boxed{\displaystyle\iint_D 2x\,dxdy = \int_0^1\int_{\frac{\sqrt{3}}{2}}^{\sqrt{1-\frac{x^2}{4}}} 2x\,dy\,dx}$ Answer: $\boxed{\dfrac{8}{3}-\dfrac{3\sqrt{3}}{2}}$.

   (c) $\displaystyle\iint_D e^{\frac{y}{x}}\,dxdy$, where D is the domain enclosed by the lines: $x=1$, $x=2$, $y=x$ and $y=2x$.

   Pay attention to the order. Answer: $\boxed{\dfrac{3}{2}(e^2-e)}$

   (d) $\displaystyle\iint_D |y-x^2|\,dxdy$, where $D := [-1,1]\times[-1,1]$.

   Hint: You might need to split $\displaystyle\iint_{D_1}|y-x^2|\,dxdy+\iint_{D_2}|y-x^2|\,dxdy$ for some domains D_1 and D_2. Answer: $\boxed{\dfrac{12}{5}}$

2. Find the area $A(D)$ of the domain

$$D := \left\{(x,y)\in\mathbb{R}^2\colon\ 3x\leq y\leq\frac{x}{3},\ x\leq 0,\ y\geq -x-3\right\}.$$

Hint: $\boxed{A(D)=\displaystyle\iint_D 1\,dxdy}$ and clearly identify the points of intersection of the lines enclosing the domain D.

The next set of notes will be on the change of variables in double integrals!